

Aryan Karki | Software Engineer | AI Automation Engineer | Data Science Student |

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Professional Summary

I am a Software Developer and Data Science student who excels in backend engineering, systems automation, and robust web applications. Most recently, I spent six months at Xuno (a US-based remittance company) building high-impact production systems: an end-to-end customer support RAG engine, a JSON-driven QA automation framework, and an OAuth-backed role-based data access layer. I thrive on solving complex backend bottlenecks, optimizing database queries, and translating modern architectural designs into highly reliable software.

Professional Experience

Xuno (Remit Platform)

Kathmandu

AI Automation & Software Engineering Intern

Nov 2025 – May 2026

Over a six-month tenure, I designed, optimized, and deployed production-grade systems across three primary technical tracks:

○ Backend Engineering & Conversational AI

- Designed and deployed an end-to-end customer-support chatbot on a verified WhatsApp Business account, scaling from a prototype to a multi-agent RAG system serving live users.
- Replaced a legacy PostgreSQL fuzzy-matching implementation with Qdrant semantic vector search; implemented cosine-similarity thresholding to eliminate hallucinations and secure factual accuracy.
- Optimized relational data flows by building a PostgreSQL context-retention schema, preserving per-user conversation states and reducing API latency.
- Architected multi-agent pipelines using n8n to orchestrate asynchronous sub-agents handling live exchange-rate API calls, rate calculations, and dynamic FAQ routing.
- Researched and benchmarked workflow engines (n8n, Zapier, Make), delivering a comparative technical analysis adopted as the engineering team's structural standard.

○ Full-Stack Tooling & QA Automation

- Built a production-grade Selenium + Python test automation framework from scratch using a CLI, Page Object Model (POM), and fully JSON-driven execution logic (18/21 core cases passing on first release).
- Expanded the test engine to support 24+ assertion and navigation types, refactoring legacy hard-coded scripts into a single, clean JSON-driven execution runner.
- Resolved complex headless Chrome driver session failures, creating stable, automated test runs compatible with continuous integration (CI) environments.
- Constructed an MCP-based developer tool connected to Claude Desktop to automate localized test orchestration using local LLMs (Qwen 2.5-Coder and Gemma 2 via Ollama).

○ DevOps, Security & Infrastructure

- Developed a Client-Side Field-Level Encryption (CSFLE) Proof of Concept for a secure MongoDB financial ledger migration, protecting sensitive transactional fields with automated dry-run verification.
- Implemented a secure OAuth 2.0 backend integrated with an MCP server to enforce role-based access control (RBAC) over sensitive database schemas.
- Configured CI/CD integrations between GitHub and Jira, using webhook automation to advance issue tickets dynamically upon pull request creation and code merges.

Projects

Xuno Placement

Kathmandu

WhatsApp RAG Support Chatbot

2025 – 2026

- **Tech Stack:** Llama 3.1 (Groq API), Qdrant Vector DB, Gemma Embeddings, n8n, PostgreSQL.
- An intelligent assistant orchestrating semantic search queries and third-party financial calculation APIs with strict state management and multi-agent routing.

Xuno Placement

Kathmandu

JSON-Driven QA Automation Framework

2025 – 2026

- **Tech Stack:** Python, Selenium, JSON, MCP Server, Ollama.
- An extensible, code-free automation runner featuring 24 configurable assertion and state interaction types.

Xuno Placement

Kathmandu

MongoDB CSFLE Ledger Migration Tool

2026

- **Tech Stack:** MongoDB, Node.js/Python, Encryption Key Vaults.
- A client-side encryption utility built to safely migrate legacy databases while ensuring PII and ledger fields remain encrypted end-to-end.

Academic

Kathmandu

Wingsten Furniture E-Commerce Website

2023

- **Tech Stack:** PHP, MySQL, JavaScript, HTML5, CSS3, Bootstrap.
- A fully responsive e-commerce web application featuring optimized relational search indexes and category browsing.

Technical Skills

Languages & Frameworks: Python, JavaScript (ES6+), PHP, HTML5, CSS3, React.js, Tailwind CSS, Flask, Django.

Databases & Vector Stores: PostgreSQL, MongoDB (CSFLE), Qdrant, pgvector.

Automation & Testing: Selenium WebDriver, n8n, Twilio, Page Object Model (POM).

System Integrations: REST APIs, OAuth 2.0, Webhooks, Model Context Protocol (MCP), LangGraph.

DevOps, Cloud & Tools: Git, GitHub, Docker (self-hosting n8n), Jira, Cloudflare Tunnels, Pandas, NumPy.

Education

Sunway College (Birmingham City University)

Kathmandu, Nepal

BSc (Hons) Computer Science with AI

2023 – Present

Relevant Coursework: Machine Learning, Database Systems, Web Technologies, Software Engineering.

Southwestern State College

Kathmandu, Nepal

Higher Secondary Certificate

2019 – 2021